Udemy ML, DL, NLP Course
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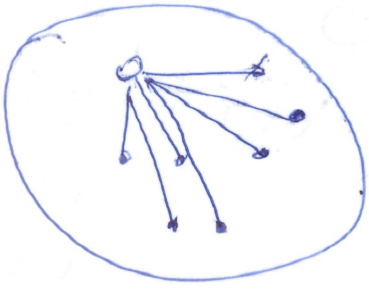
Lec - Silhouette Clustering

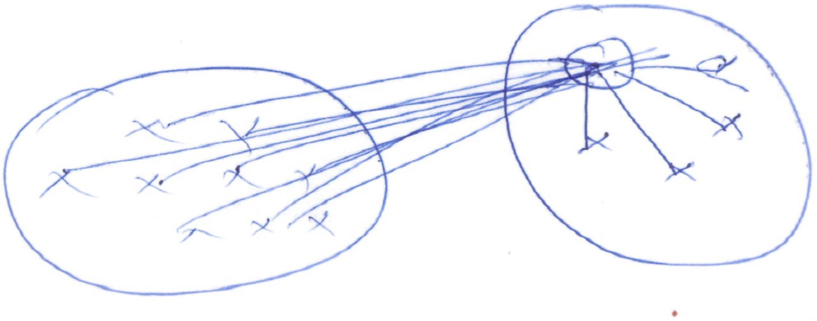
Intution of silhouette clustering

K means
K = 4 Elbow method

Classification → Model

⇒ $a(i)$





$a(i) << b(i$

we now define a silhouette of one data point i

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

if $|C_i| > 1$

and

$s(i) = 0$, if $|C_i| = 1$

It can also be written as

$$s(i) = \begin{cases} 1 - a(i)/b(i), & \text{if } a(i) < b(i) \\ 0, & \text{if } (a(i) = b(i) \\ b(i)/a(i) - 1 & \text{if } \cancel{(a)}(i)\ a(i) > b(i) \end{cases}$$

(-1 to 1)

max ~~loss~~ near to 1
better clustering model
has to be created